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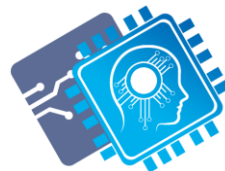
BOLD-Q: Blockwise Outlier-aware Logarithmic Dual-Bias Quantization for Hardware-Efficient LLM Inference

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IDS Lab

Intelligent Digital Systems Design Lab.



- I . Motivation & Contribution**
- II. Backgrounds**
- III. Proposed Methods**
- IV. Experimental Results**

Motivation & Contribution



- **Goal:** A hardware-efficient, **outlier-aware quantization framework that combines the logarithmic number system (LNS) and Microscaling (MX)** for LLM Inference
- **Motivation:** MX quantization yields a homogeneous execution path across Transformer blocks—compact for on-device AI; However, **low-precision MX quantization** has the following limitations: **fixed bins (L1), coarse shared-exponent (L2), accuracy–efficiency gap across existing methods (L3)**
- **Solution/Contribution:**
 - 1) **LNS base Dual-Bias** : Use a per-block Dual-Bias to adaptively reshape bins and accommodate blocks skewed by outliers
 - 2) **LNS Scaling** : Apply a finer-grained LNS-based scaling factor than shared-exponent scaling to better match per-block distributions
 - 3) **Systolic Array** : Deploy an adder-based LNS–MAC systolic array, improving compute and memory efficiency for on-device execution

Table. 1. Comparison of LLM quantization methods across key design criteria

LLM Quant.	ANT [1]	Tender [2]	MXFP [3]	AMXFP [4]	BOLD-Q (ours)
Quant. method	Calibration base static Quant.		Dynamic Block Quant.		
W/O Calibration	X	X	✓	✓	✓
Attention Quant.	X	X	✓	✓	✓
HW Efficiency	MEDIUM	HIGH	HIGH	LOW	HIGH
High Accuracy	X	X	X	✓	✓

- [1] “Ant: Exploiting adaptive numerical data type for low-bit deep neural network quantization,” in *Proc. IEEE/ACM MICRO*, Oct. 2022.
- [2] “Tender: Accelerating large language models via tensor decomposition and runtime requantization,” in *Proc. ACM/IEEE ISCA*, Jun. 2024.
- [3] “Microscaling data formats for deep learning,” *arXiv:2310.10537*, 2023.
- [4] “AMXFP4: Taming activation outliers with asymmetric microscaling floating-point for 4-bit LLM inference,” in *Proc. ACL Findings*, Jul. 2025.

Background

- **Microscaling (MX)**

- Fine-grained granularity
 - Tensor, Channel, **Block**(e.g., # of 32 data)
- **Shared-exponent Scaling**
 - **Easy to dynamic dequantization/quantization**
- Inferior Performance on low precision
 - **Quantization error via shared exponent**

- **Logarithmic Number System(LNS)**

- **LNS value representation**
 - $x = (-1)^s 2^y$
 - a real number x
 - a sign bit s and its logarithmic magnitude y
- **Format Example ($LNS8_{E4M3}$)**
 - Consists of 1 sign, 4 exponent, 3 fractional bits
 - The logarithmic value y adopts a fixed-point representation
 - for example, $x = 2^{2.75} \rightarrow y = 2.75(00010.110_2)$
- **In the LNS domain, multiplication reduces to addition**
 - $P = \log_2(a \times b) = \log_2 a + \log_2 b = y_a + y_b$
 - a, b are positive real numbers

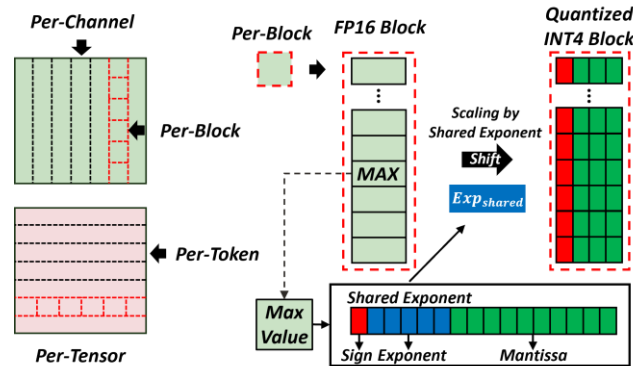


Fig. 1. Microscaling quantization scheme

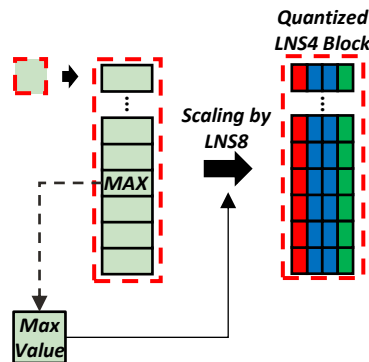


Fig. 2. LNS scaling base MX quantization to LNS4

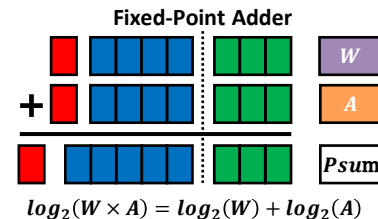


Fig. 3. Multiplication in LNS domain

Proposed Method (1/2)

- **Solution/Contribution**
 - Propose an outlier robust quantization framework that introduces **the LNS-based scaling factor and Dual-Bias**
- **Quantization Framework of BOLD-Q**
 - **LNS Based Scaling**
 - **Apply a finer-grained LNS-based scaling**
 - No additional overhead thanks to adder-based LNS operations
 - **Solution for L2. Coarse Scaling Granularity**
 - **Dual-Bias (1)**
 - Offset composed of LNS 4-bit Sub-Bias and Bias
 - **Be injected into the top two bin values of $LNS4_{E2M1}$**
 - **Solution for L1. Insufficient Resolution**
 - **Offline Block-Specific Dual-Bias for Weight Quant. (OBWQ) (2)**
 - Performs a static, per-block search for the Dual-Bias
 - Evaluate the mean square error (MSE) over all candidate pairs
 - **Adaptive Dual-Bias Quant. (ADBQ) (3)**
 - Unlike weights, search-based methods are impractical at runtime
 - Extract the top-1 to top-3 magnitudes from each block
 - Apply the Dual-Bias computation algorithm

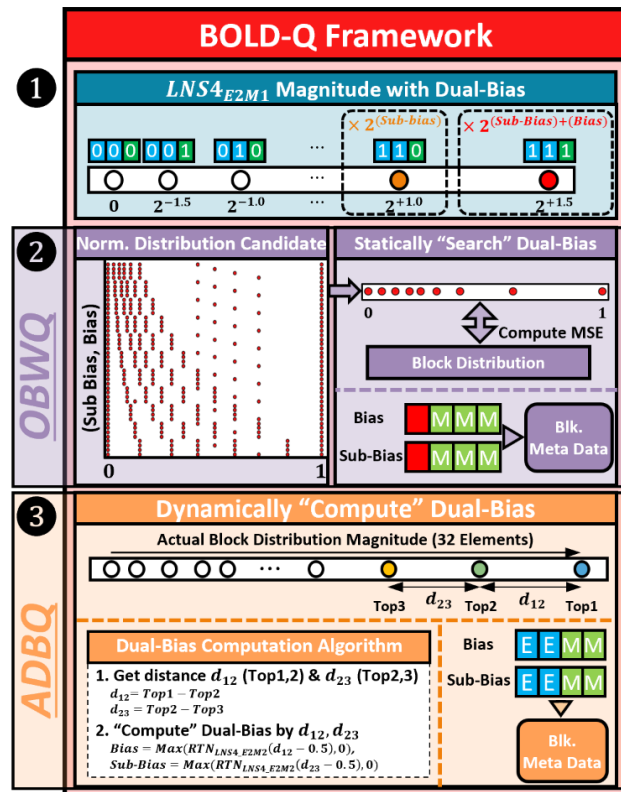


Fig. 4. Framework of BOLD-Q

- ### BOLD-Q Architecture

S Sign
 E Exponent
 M Fraction
 W Weight
 A Activation
 Psum Partial sum

(a) W , Dual-Bias, LNS scale factor

(b) Dual-Bias Preprocessing

(c) LNS-MAC PE

 - 8-bit Fixed-point Adder
 - LUT base Accumulation

Fixed-Point Adder									
	E	E	E	E	M	M	M	W	
+	E	E	E	E	M	M	M	A	
	E	E	E	E	M	M	M	Psum	

$\log_2(W \times A) = \log_2(W) + \log_2(A)$

(d) Encoding to LNS16

(e) Scaling by LNS scale factor

(f) "Computing" Dual-Bias

(g) Quant. Buffer

(h) Weight Buffer

(i) Bias Gen.

(j) Fixed-point Adder

(k) Align

(l) Fixed-point Adder

(m) LUT

(n) Fixed-point Adder

(o) Dequant. Module

(p) Quant. Module

(q) ADBQ

(r) Fixed-point Adder

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Experimental Results

Algorithm/Hardware Performance Analysis

- BOLD-Q** achieves **superior perplexity performance compared with competing research** across multiple bit-precision configurations
- BOLD-Q** outperforms Quarot [9] in perplexity **even under lower bit-precision settings in the attention layer** of the LLaMA 2 7B model
- Moreover, **BOLD-Q** achieves **34.0% area and 21.4% energy reductions** compared to AMXFP [4]

Table 2. Quantization Results for Various LLMs on WikiText-2 (perplexity ▼)

Method	Linear (W/A)	Atten. (W/A)	Scale Factor	Block Size	OPT 6.7B	OPT 13B	LLaMA1 7B	LLaMA1 13B	LLaMA2 7B	LLaMA2 13B	LLaMA3 8B	Mistral 7B
Baseline	16/16	16/16	-	-	10.86	10.13	5.68	5.09	5.47	4.88	6.14	5.25
GPTQ [5]	4/16	16/16	FP16	128	10.93	10.17	5.83	5.20	5.63	4.99	6.56	5.39
AWQ [6]	4/16	16/16	FP16	128	10.93	10.21	5.78	5.19	5.60	4.97	6.54	5.37
OmniQuant [7]	4/16	16/16	FP16	128	10.96	10.20	5.77	5.17	5.58	4.95	-	-
AMXFP [4]	4/16	16/16	FP8	32	10.96	10.34	5.84	5.21	5.64	4.98	-	5.39
BOLD-Q (ours)	4/16	16/16	LNS8	32	10.89	10.16	5.76	5.15	5.55	4.95	6.40	5.32
ANT [1]	8/8	16/16	FP16	-	19.72	4E+3	8.52	7.49	8.79	20.52	-	-
QServe [8]	4/8	4/16	FP16	128	-	-	5.89	5.25	5.70	5.08	6.76	5.42
MXFP [3]	4/8	4/8	PoT8	32	12.27	11.64	6.81	5.91	6.75	6.01	-	5.88
AMXFP [4]	4/8	4/8	FP8	32	11.40	10.99	5.99	5.35	5.85	5.20	-	5.48
BOLD-Q (ours)	4/8	4/8	LNS8	32	11.20	10.74	5.97	5.28	5.79	5.14	6.70	5.41
ANT [1]	4/4	16/16	FP16	-	9E+3	4E+4	80.13	96.71	189.72	165.19	-	-
Quarot [9]	4/4	4/16	FP16	Channel	-	-	6.34	5.58	6.10	5.40	8.17	5.80
ATOM [10]	4/4	4/16	FP16	128	-	-	6.16	5.46	6.12	5.31	7.76	5.76
MXFP [3]	4/4	4/4	PoT8	32	22.51	12.88	10.20	7.80	11.18	6.98	11.17	9.43
AMXFP [4]	4/4	4/4	FP8	32	13.06	11.90	6.25	5.52	6.22	5.47	7.72	5.71
BlockDialect [11]	4/4	4/4	PoT8	32	11.63	-	6.39	-	6.33	-	7.87	5.74
BOLD-Q (ours)	4/4	4/4	LNS8	32	11.45	10.97	6.18	5.45	6.07	5.37	7.34	5.56

Entries with unreported or unreproducible results are marked “-”.

Table 3. Area for BOLD-Q and baseline architectures

Arch.	Core		Others		
	Component (μm^2)	Num.	Area (mm^2)	Dequant. (mm^2)	Quant. (mm^2)
AMXFP	PE (521.35)	1024	0.542	0.03	0.039
	Encoder (243.83)	32			
MXFP	PE (443.20)	1024	0.458	0.003	0.008
	Encoder (121.91)	32			
ANT	PE (423.77)	1024	0.432	0.017	0.017
	Decoder (12.64)	32			
	Encoder (121.91)	32			
BOLD-Q (Ours)	PE (372.41)	1024	0.383	0.003	0.017
	Preproc. (89.97)	32			
	Encoder (128.00)	32			

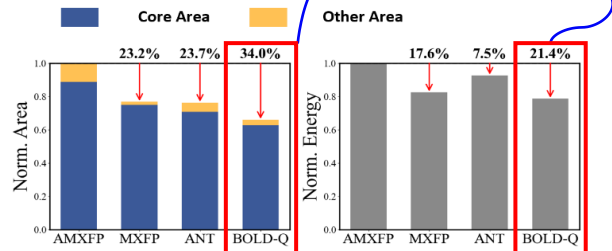


Fig. 6. Comparison of normalized area and energy 7

[1] ANT: Exploiting Adaptive Numerical Data Type for Low-Bit Deep Neural Network Quantization, MICRO, 2022 / [3] Microscaling Data Formats for Deep Learning, arXiv, 2023
 [4] AMXFP4: Asymmetric Microscaling Floating-Point for 4-bit LLM Inference, ACL Findings, 2025 / [5] GPTQ: Accurate Post-Training Quantization for Generative Pre-Trained Transformers, ICLR, 2023
 [6] AWQ: Activation-Aware Weight Quantization for LLM Compression and Acceleration, MLSys, 2024 / [7] OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models, ICLR, 2024
 [8] QServe: W4A8KV4 Quantization and System Co-Design for Efficient LLM Serving, MLSys, 2025 / [9] Quarot: Outlier-Free 4-Bit Inference in Rotated LLMs, NeurIPS, 2024
 [10] ATOM: Low-bit Quantization for Efficient and Accurate LLM Serving, MLSys, 2024 / [11] BlockDialect: Block-wise Fine-grained Mixed Format Quantization for Energy-Efficient LLM Inference, arXiv, 2025

THANK YOU!